

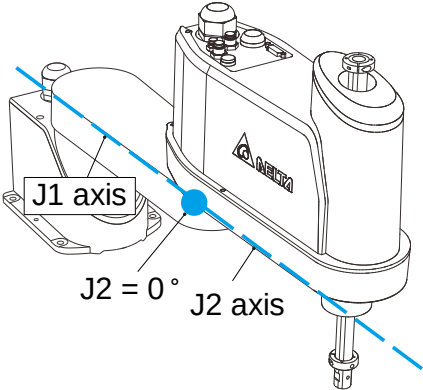
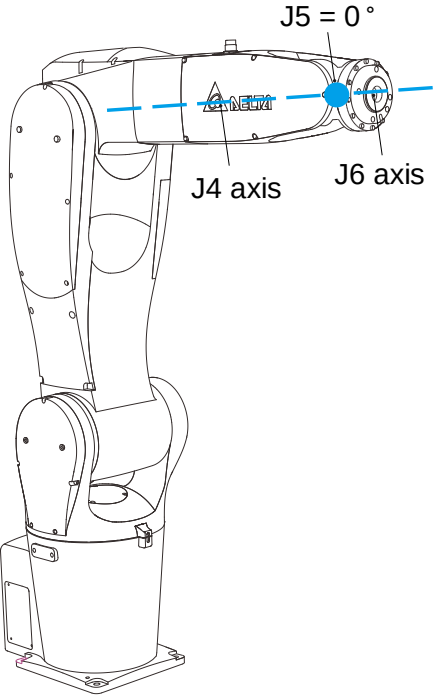
Preface

This manual is written based on Delta Robotic Automation Studio (DRASudio) V1.00.07.55. Check for the version number before operating the software. If the version number is below V1.00.07.55, contact Delta and the distributor before operation.

Safety precautions

Singularities

Both SCARA and VA robots have singularities. The following describes singularity occurrence and avoidance.

SCARA singularity	VA singularity
 The diagram shows a SCARA robot arm in a side view. A blue dashed line represents the J1 axis, and another blue dashed line represents the J2 axis. The two axes are shown as coincident, with a label 'J2 = 0°' indicating the singularity condition. The J1 axis is labeled 'J1 axis' and the J2 axis is labeled 'J2 axis'.	 The diagram shows a VA robot arm in a side view. A blue dashed line represents the J4 axis, and another blue dashed line represents the J6 axis. The two axes are shown as coincident, with a label 'J5 = 0°' indicating the singularity condition. The J4 axis is labeled 'J4 axis' and the J6 axis is labeled 'J6 axis'.
Singularity occurrence: During robot movement, $J2 = 0^\circ$ and the J1 axis and J2 axis become coincident. At this point, the J2 axis stops and cannot move. How to avoid singularities: ■ Move J2 by $\pm 5^\circ$ to 15°. ■ Add multiple joint motions to avoid J2 from being 0°. ■ Use the MovP instructions.	Wrist singularity occurrence: During robot movement, $J5 = 0^\circ$ and the J4 axis and J6 axis become coincident. At this point, the J4 axis instantly rotates at a high speed; beware that the cables on the J4 axis are pulled along with the rotation. How to avoid singularities: ■ Move J5 by $\pm 5^\circ$ to 15°. ■ Add multiple joint motions to avoid the J4 axis and J6 axis from becoming coincident. ■ Use the MovP instructions.